Kafrelsheikh University

Faculty of Medicine

Medical Microbiology &Immunology

Department

Innate Immunity

By

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Learning outcomes

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By the end of the lecture, the students will be able to:

- Realize types and function of the immune system
- Describe different mechanisms of innate immunity
- What are the immune system cells
- Define macrophages and their function.

Lecture Outline

- What are the two types of immune system
- What are the immune system cells
- macrophages and their function.

Immunity

- Means resistance, protection and non-susceptibility to invasion by foreign substances

- **The main function of the immune system is to prevent or limit infections by microorganisms such as bacteria, viruses, fungi and parasites.**

- **This defensive function is performed by various cellular and humoral components which interact with each other producing a coordinated immune response that eliminates the pathogen or minimizes the damage it causes.**

Functions of immune response

	1. Defense	2. Homeostas (Immunological balance)	3.Surveillance (Immunological control)
(A) Normal Condition	Resistance to infection by organisms.	Removal of damage components.	Recognition and destruction of abnormal cell types (mutants) which arise within the body from time to time.
(B) Aberration with hyper-function	Allergy	Autoimmune diseases.	———
(C) Aberration with hypo-function	Increase the susceptibility to repeated infections	———	Malignant diseases .

The immune system is divided into two functional divisions:

1) Natural (innate or non-specific) immune response
2) Adaptive (acquired or specific) immune response

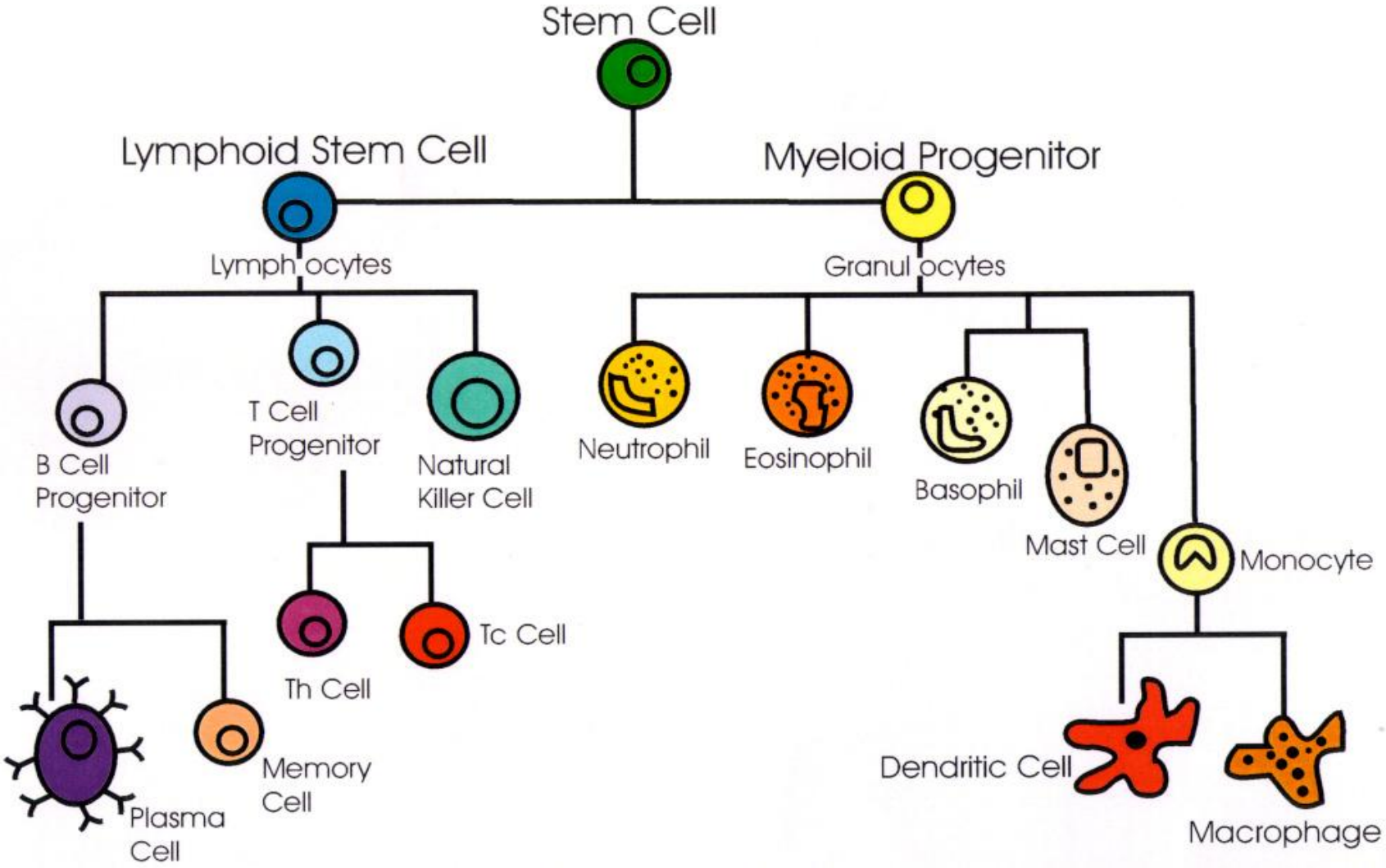
- Innate immunity acts as a first line of defense against infectious agents .
- If these innate defenses fail to eliminate the pathogen ,the adaptive immunity will be activated to eliminate this pathogen
- This immune response is then retained after the pathogen has been eliminated, in the form of an immunological memory, and allows the adaptive immune system to mount faster and stronger attacks each time this pathogen is encountered
- The innate and acquired immunity do not operate in total independence of each other. They cooperate in important ways to produce more effective immunity.
- The communication system through various cytokines and adhesion molecules, allow components of innate and acquired immunity to interact, send each other signals, activate each other and work in concert towards the final goal of destroying the invading microorganism.

Innate (non-specific or natural) immunity:

The characteristics of the innate immune response include the following:

- Acts **immediately** as the first line of defense against infectious agents
- Responses are Broad-Spectrum (**non-specific**)
- There is **no memory** or lasting protective immunity
- Such mechanisms are **present since birth**

Cells of the Immune System



A) Cells involved in innate immunity

1) Phagocytic cells (Monocyte, Macrophage, Neutrophil):

- **Monocyte**: The largest nucleated cell of the blood, developing into a macrophage when it migrates into the tissues.
- **Macrophage**: The main phagocytic cells fixed cells in tissues and serous cavities such as the pleura and peritoneum. They have different names in different tissues e.g. Kupffer cells in liver, **they are important link between innate and acquired immune responses**
- **Neutrophil**: also known as polymorphonuclear leukocytes, the commonest leukocytes of the blood, a short-lived phagocytic cell whose granules contain numerous bactericidal substances.

2) Inflammatory cells (Mast cells, Eosinophils, Basophils):

- **Mast cells:** Round central nucleus surrounded by large darkly stained granules. They reside in connective tissues and mucous membranes. They are most often associated with allergy and anaphylaxis by releasing histamine
- **Eosinophils:** cells with multi-lobed nuclei and eosinophilic (red) cytoplasmic granules. They are involved in allergic reactions and killing of larger parasites including helminthes.
- **Basophils:** cells with heavily stained blue (basophilic) granules that contain heparin and vasoactive amines, important in the inflammatory response. They also secrete the chemicals responsible for immediate hypersensitivity

3) Natural Killer Cells (NK cells): a small, distinct group of large granular lymphocytes. They represent a first line of defense to infections, tumor growth and other pathogenic alterations of tissue homoeostasis because they non-specifically kill any cancerous or virus-infected cells by releasing perforins & cytolytic chemicals. They do not express antibodies or T cell receptors at their cell surface. MHC I inhibits their killing functions.

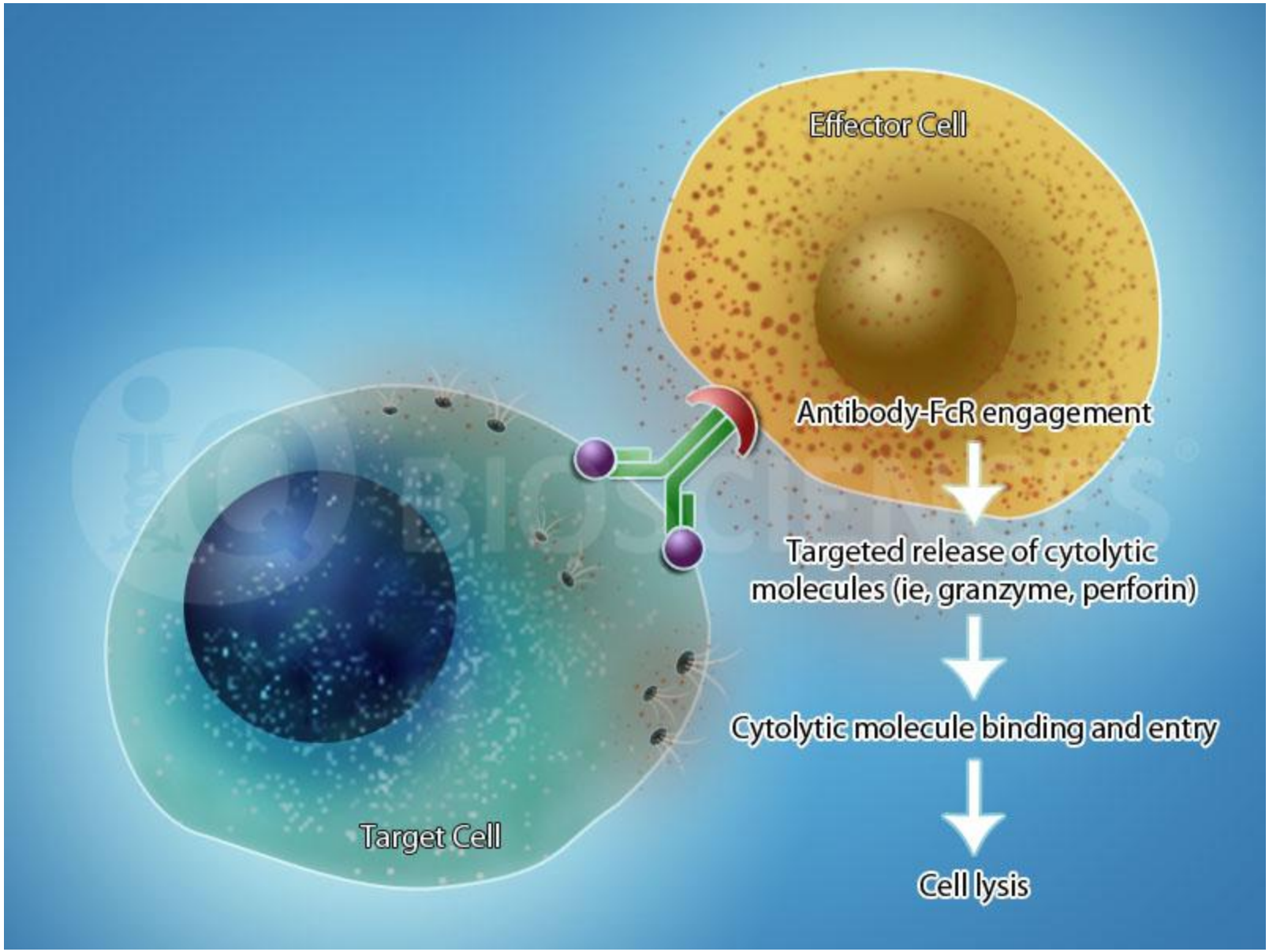
NK cells differ from CTLs in the following;

1-They are non-specific i.e. one NK cell can kill many different abnormal cells.

2-They act spontaneously without prior recognition or activation.

3-They do not require antigen presentation by MHC i.e. not MHC restricted.

4-They destroy cells coated with antibodies, a mechanism called antibody dependent cellular cytotoxicity (ADCC). NK cells have Fc receptors for IgG, which bind to antibody coated cells and kill these cells.


Effector Cell
Antibody-FcR engagement
Targeted release of cytolytic molecules (ie, granzyme, perforin)
Cytolytic molecule binding and entry
Cell lysis
Target Cell

Mechanisms of innate immunity:(imp)

I) Anatomic & mechanical barriers.

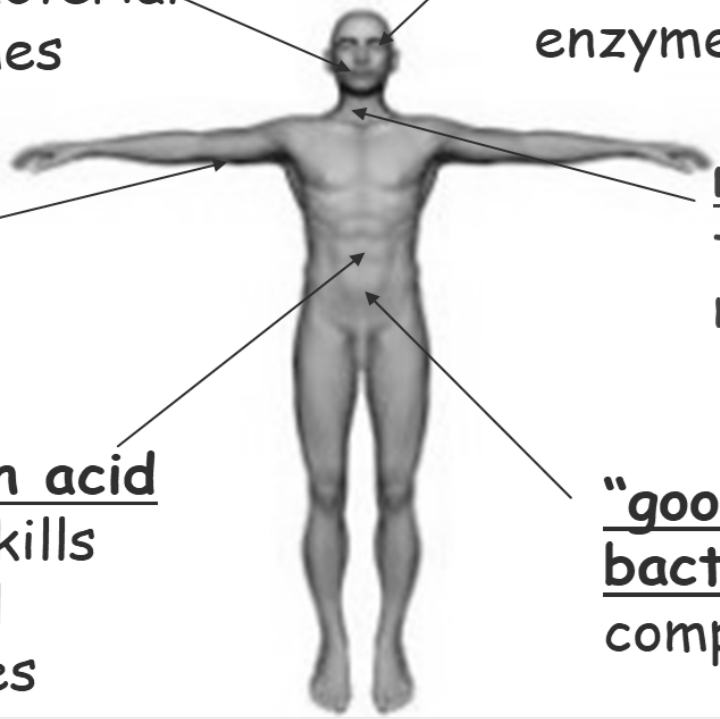
II)Chemical barriers.

III) Inflammatory barriers

IV) Fever

V) Phagocytic barriers.

First lines of defence



saliva
antibacterial enzymes

tears
antibacterial enzymes

mucus linings traps dirt and microbes

skin
prevents entry

stomach acid
low pH kills harmful microbes

"good" gut bacteria out compete bad

I) Anatomic & mechanical barriers at portal of entry .

- **The intact skin and mucous membranes** :They are effective barriers against most microorganisms.
- **Epithelia** of the skin, GIT, and respiratory tract produce peptides that have natural antibiotic functions e.g. defensins
- **The hair** at the nares, coughing and sneezing help to expel foreign particles.
- **Mucous secretions** trap many organisms which are pushed outside the body by the continuous movement of cilia in the respiratory tract.
- **The blinking reflex and tears** expel foreign particles or bacteria entering the conjunctiva.

2) Chemical Barriers at the portal ofentry.

- **Sweat and sebaceous secretions** in the skin have antimicrobial actions by virtue of their acidic pH and high fatty acids content.
- **Hydrolytic enzymes** in the saliva, **HCL** of the stomach, **proteolytic enzymes** in the small intestine are bactericidal.
- **Lysozyme,** an enzyme that dissolves bacterial cell walls (by breaking down peptidoglycan). It is present on the skin, in tears and cervical secretions.
- **Acidic pH** in the adult vagina is protective.

3) Normal microbiota present at the portal of entry

- They suppress the growth of many pathogenic bacteria and fungi by competition for essential nutrients or by production of inhibitory substances such as colicins or acids.
- For example, in the adult vagina an acidic pH is maintained by normal microbiota namely lactobacilli, that interfere with the establishment of pathogenic organisms.
- Suppression of normal microbiota by antibiotics leads to superinfection with potential pathogens.

4) Circulating effector proteins in serum and body fluids :

They suppress the growth and promote killing of microbes, They include:

- **Complement**: It is a group of proteins in serum and body fluids (will be discussed later)
- **Acute phase proteins**: These are substances that increase in response to inflammation and include; C-reactive protein (CRP),Measuring CRP is a useful laboratory test to assess activity of inflammatory diseases.
- **Interferons (IFNs)**:
- **Cytokines**

5) Inflammatory Barriers:

- Tissue damage by a wound or by an invading pathogen, induces "inflammatory response".
- vasodilatation of nearby capillaries leading to redness of tissues and increase of tissue temperature,
- increased capillary permeability and influx of fluids and cells (exudates) into the tissues causing oedema,
- influx of phagocytes from the capillaries into the site of tissue damage, these engulf bacteria and release lytic enzymes that can damage nearby healthy cells forming the pus.

- A variety of chemical mediators are responsible for initiation of the inflammatory response.
- Among these chemical mediators are the acute phase proteins and cytokines.
- These participate in destroying invaders

Phagocytic barrier (Phagocytosis):

- Phagocytosis means ingestion of particulate material including whole pathogenic microorganisms.

Steps of phagocytosis:

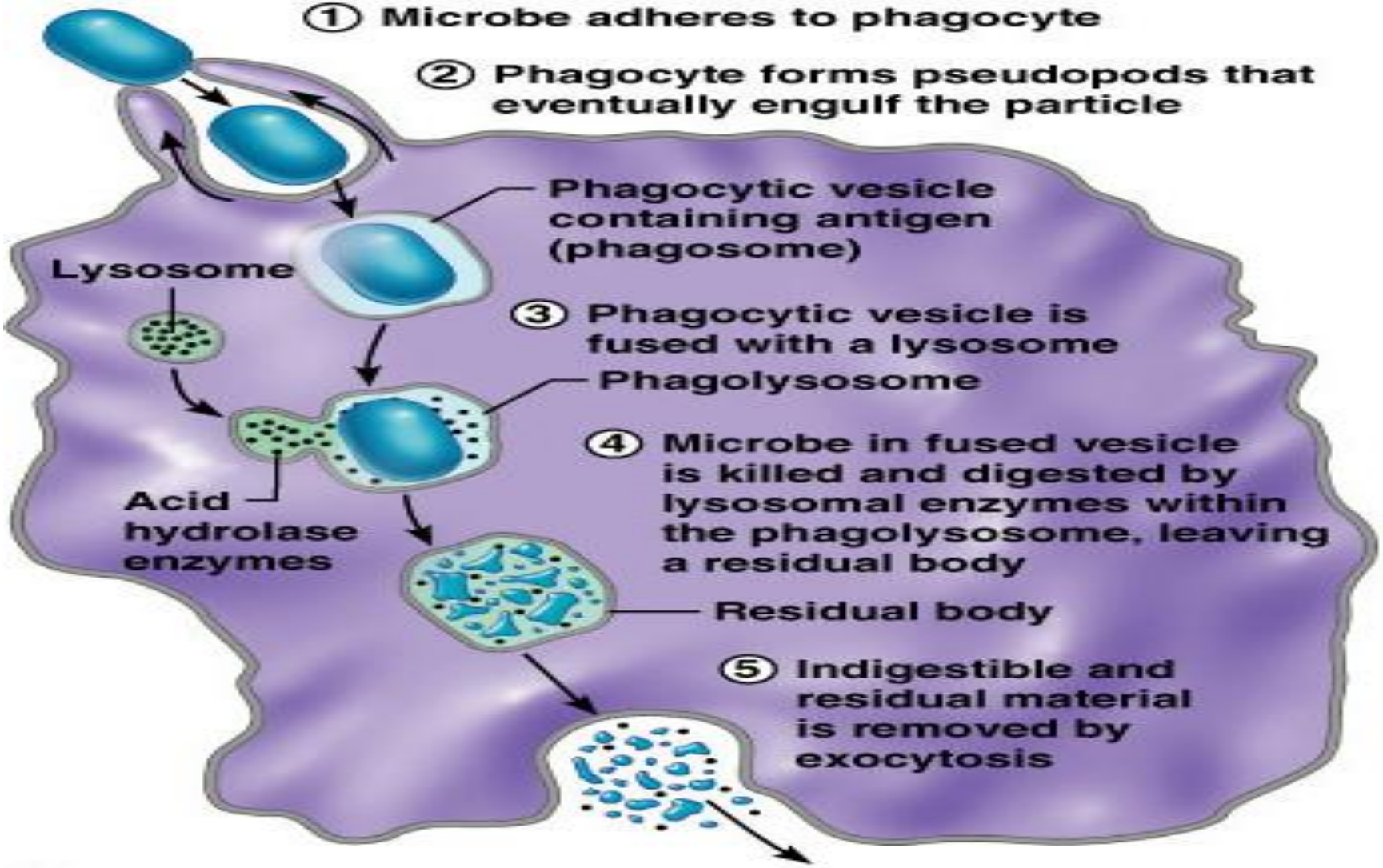
1- Chemotaxis: movement of leukocytes towards the site of infection due to chemical stimulus.

2-Binding: phagocytes adhere to foreign particles or microorganism.

3-Engulfment: ingestion of the microbe by the pseudopodia forming phagosome.

4-Phagosome fuse with intracellular lysosome (contain digestive enzymes) forming phagolysosome.

5- Intracellular Killing of foreign particles: through oxygen radicals (i.e. superoxide or hydrogen peroxide) and nitrogen radicals (nitric oxide). Then Processed antigens are extruded outside the cell by exocytosis. This is required to activate the acquired immunity i.e. this step is a link between innate and adaptive immunity.



Steps of phagocytosis:

Functions of Macrophages:

1. **Phagocytosis**
2. **Major antigen presenting cells**: They ingest foreign material, process it and fragments of antigen are presented (expressed) on their surface in association with class II antigens of the major histocompatibility complex (MHC II molecules) for interaction with T-cells.
3. Macrophages surface Fc receptors interact with Fc portion of the antibody molecule enhancing uptake of organisms coated with antibody **(opsonization).**
4. Macrophages may kill antibody coated infected cells or tumor cells by **antibody dependent cell mediated cytotoxicity (ADCC).**
5. **Release of soluble mediators:**
   - They produce interleukin-1 & interleukin-6 and tumor necrosis factor (TNF α).
   - They secrete prostaglandins and synthesize complement components.

Opsonisation: is the promotion or enhancement of phagocytosis ,

- Macrophages have surface C3b and Fc receptors that interact with Fc portion of IgG thereby enhancing the uptake of organisms coated with antibody alone or antibody and complement.
- Attachment and ingestion of microbes are greatly enhanced if the organism is coated by its specific antibody, by activated complement C3b or by antibody and C3b (opsonins) via the C3 or Fc receptor.

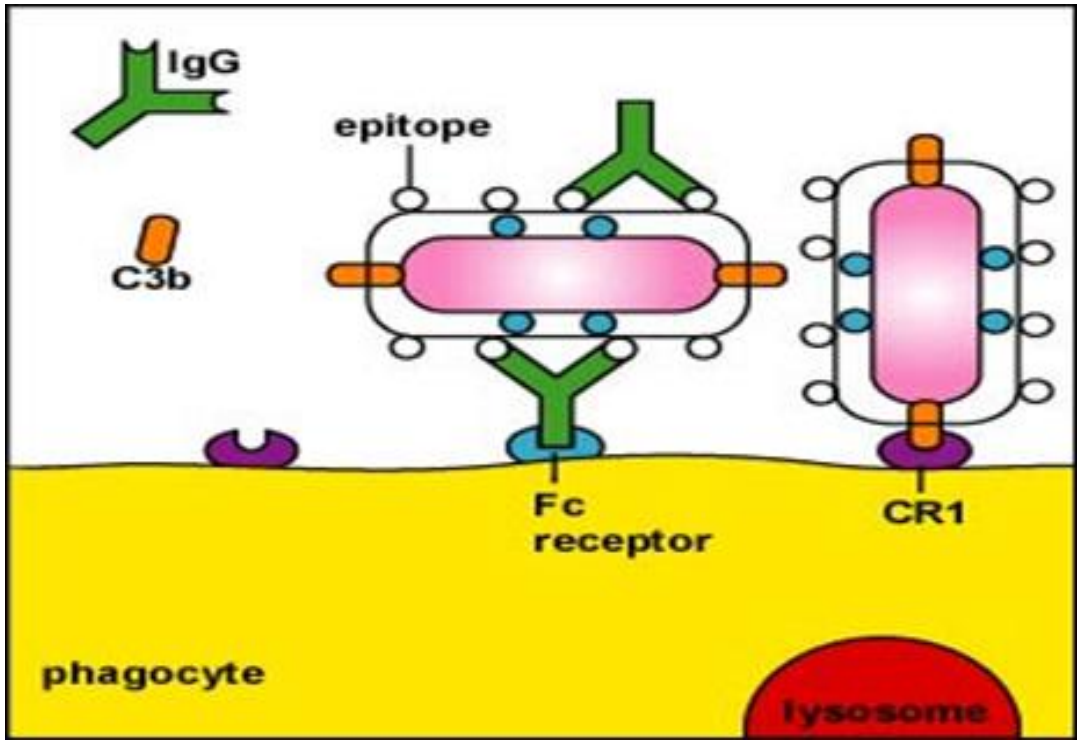
Opsonin :

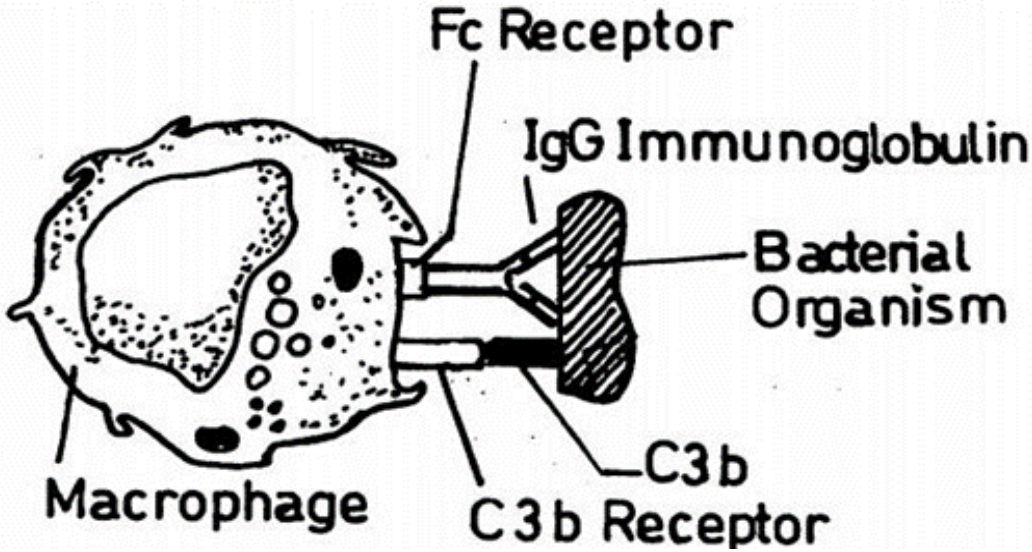
A macromolecule that becomes attached to the surface of a microbe that can be recognized by the surface receptors of neutrophils and macrophages and that increases the efficiency of phagocytosis of the microbe.

Types of opsonins :

a) Heat labile (C3b complement component).

b) Heat stable (IgG immunoglobulins).


IgG
epitope
C3b
Fc
receptor
CR1
phagocyte
lysosome


Fc Receptor
IgG Immunoglobulin
Bacterial
Organism
C3b
Macrophage
C3b Receptor

Movement of cells in response to a chemical stimulus is called:

a) Addressinazition.
b) Chemotaxis.
c) Diapedisis.
d) Marginalization.
e) Opsonization.

- **Opsonins include which one of the following?**

a. Perforin

b. C5a

c. C9

d. IFNγ

e. C3b

Which of the following is not characteristic of the innate immune response ?

a) Acts **immediately** against infectious agents

b) Responses are Broad-Spectrum (**non-specific**)

c) There is **memory** or lasting protective immunity

d) Such mechanisms are **present since birth**

Summary

- There are two types of immune response
- There are different mechanisms of innate immunity including physical and chemical barriers ,cells, circulating protiens and inflammatory response
- Steps of Phagocytosis
- Definition and mechanism of opsonozation

References

- Basic Immunology : functions and disorders of the immune system , fifth edition ; Abul K. Abbas, Andrew H. Lichtman and Shiv Pillai
- Immunology :7th edition ; David Male, Jonathan Brostoff, David Roth and Ivan Roitt



Thanks for your attention